Step 1:

Here, less days means less pay. So, it is in direct proportion.

$\therefore$ The proportion is $20 : 12 :: 6 : x$ $\longrightarrow$ (1)

Step 2:

Also, less workers means less pay. So, it is in direct proportion again.

$\therefore$ The proportion is $15 : 5 :: 6 : x$ $\longrightarrow$ (2)

Step 3:

Combining (1) and (2),

$$\left.\begin{matrix} 20 : 12 \\ 15 : 5 \end{matrix}\right\} :: 6 : x$$

We know that, **the product of the extremes = the product of the means**

Extremes		Means		Extremes
20	:	12 : 6	:	x
15	:	5		

So, $20 \times 15 \times x = 12 \times 6 \times 5 \Rightarrow x = \dfrac{12 \times 6 \times 5}{20 \times 15}$ = ₹ 1.2 lakh.

Multiplicative Factor Method

Workers	Pay (Work)	Days
15	6	20
D	D	D
5	x	12

Here, the unknown is the pay (x). It is to be compared with the workers and the days.

Step 1:

Here, less days means less pay. So, it is in direct proportion.

$\therefore$ The multiplying factor is $\dfrac{12}{20}$ (take the reciprocal).

Step 2:

Also, less workers means less pay. So, it is in direct proportion again.

$\therefore$ The multiplying factor is $\dfrac{5}{15}$ (take the reciprocal).

Step 3:

$\therefore x = 6 \times \dfrac{12}{20} \times \dfrac{5}{15}$ = ₹ 1.2 lakh.

Formula Method

Here, $P_1 = 15, D_1 = 20$ and $W_1 = 6$. Also, $P_2 = 5, D_2 = 12$ and $W_2 = x$

Using the formula, $\dfrac{P_1 \times D_1}{W_1} = \dfrac{P_2 \times D_2}{W_2}$

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